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Department of Computer Engineering
Lab Assignment

Title: Comparative Analysis of Classical Symmetric Ciphers

1. Aim

To implement classical symmetric encryption algorithms and compare their characteristics based on **confusion, diffusion, avalanche effect, frequency distribution concealment, key space, and encryption methodology**, thereby understanding the evolution of symmetric encryption techniques.

2. Learning Outcomes

After completing this experiment, students will be able to:

- Implement classical symmetric encryption algorithms.
- Perform encryption and decryption using different key generation techniques.
- Compare ciphers based on cryptographic design principles.
- Analyze the effects of confusion and diffusion.
- Evaluate the statistical characteristics of ciphertext.
- Explain how classical ciphers influenced modern symmetric cryptography.

3. Software Requirements

- Python 3.x
- Matplotlib
- Pandas (optional)
- NumPy (optional)

4. Dataset

Each student will be provided with
classical_cipher_assignment_dataset.csv
The dataset contains **100 plaintext messages**.
Each record contains

- Plaintext
- Key information for each cipher
- Empty columns to record ciphertext
- Empty columns to record observations

Students **must not modify** the plaintext.
Students are required to populate all blank columns after completing the experiment.

5. UID-Based Key Generation

Every student shall generate **unique keys** using their **10-digit UID**.
Example
UID = 2024300020

5.1 Caesar Cipher Key

Compute: Caesar Key = (Sum of all UID digits mod 25) + 1

Example: 2+0+2+4+3+0+0+0+2+0 =13

Caesar Key = 14

Possible values: 1–25

5.2 Monoalphabetic Cipher Key

Use the UID as the seed of the random number generator.

random.seed(int(UID))

Shuffle

ABCDEFGHIJKLMNOPQRSTUVWXYZ

The shuffled alphabet becomes the substitution key.

Example

Plain Alphabet

ABCDEFGHIJKLMNOPQRSTUVWXYZ

↓

Generated Alphabet

QJMYO...

The generated alphabet must be recorded in the report.

5.3 Vigenère Key

Split the UID into five groups of two digits.

Example

20 24 30 00 20

For each group

Letter

= (Number mod 26)

A=0

B=1

...

Z=25

Example

Group	Value	Letter
20	20	U
24	24	Y
30	4	E
00	0	A
20	20	U

Keyword

UYEAU

5.4 Playfair Key

Generate the Playfair keyword using the **same Vigenère keyword**.

Remove repeated letters while preserving order.

Example
UYEAU
↓
UYEA
Construct the Playfair matrix using this keyword.

5.5 Hill Cipher Key

Use the first four digits of the UID.
Example
2024
Construct
|20 24|
|30 02|
Reduce modulo 26.
|20 24|
| 4 2|
If the matrix is **not invertible modulo 26**, repeatedly add **1 (mod 26)** to the bottom-right element until gcd(det,26)=1
The resulting matrix becomes the Hill key.
This guarantees that every student obtains a valid Hill cipher key.

6. Experimental Procedure

For every plaintext in the dataset

Step 1

Generate all encryption keys using your UID.

Step 2

Encrypt every plaintext using

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher
- Playfair Cipher
- Hill Cipher

Populate

Step 3

Decrypt every ciphertext.
Verify: Decrypt(Encrypt(Plaintext)) = Plaintext

7. Experimental Analysis

The objective of this experiment is to compare the ciphers based on their security characteristics.

Part A – Confusion

Choose one plaintext from the dataset.
Generate a **second set of keys** by changing the last digit of your UID by **+1 modulo 10**.
Encrypt the same plaintext again.
Observe

- How much does the ciphertext change?

- Does changing the key significantly affect the ciphertext?

Part B – Diffusion

Choose one plaintext.

Modify **only one character**.

Example

NETWORK SECURITY

↓

NETWORK XECURITY

Encrypt both messages.

Count

Number of ciphertext characters that changed

Part C – Avalanche Effect

$$\text{Avalanche Effect } (\%) = \left(\frac{\text{Changed Ciphertext Characters}}{\text{Total Ciphertext Characters}} \right) \times 100$$

Part D – Frequency Distribution Concealment

Select the longest plaintext in the dataset.

For every algorithm

1. Encrypt the plaintext.
2. Plot the frequency distribution of the plaintext.
3. Plot the frequency distribution of the ciphertext.

Answer

- Which algorithm preserves frequency distribution?
- Which algorithm conceals statistical information most effectively?

Part E – Key Space

Estimate the effective key space.

Part F – Comparative Evaluation

Based on your observations, complete the following table.

Property	**Caesar**	**Mono**	**Vigenère**	**Playfair**	**Hill**
Type of Cipher					
Unit of Encryption					
Confusion					
Diffusion					
Frequency Concealment					
Avalanche Effect					
Key Space					
Mathematical Operations					
Overall Security					

8. Discussion Questions

1. How does increasing the key space improve security?
2. Why is confusion an important property of an encryption algorithm?
3. Which algorithm provides the greatest diffusion? Explain using your observations.
4. Why does the Hill cipher demonstrate better diffusion than substitution ciphers?
5. Which algorithm best conceals the frequency distribution of plaintext?
6. How do the concepts of confusion and diffusion introduced by classical ciphers influence the design of DES and AES?

9. Submission

Students shall submit:

1. Python source code for all five ciphers.
2. The completed classical_cipher_assignment_dataset.csv.
3. Frequency distribution plots.
4. Completed comparative evaluation tables.
5. Answers to the discussion questions.
6. A one-page conclusion summarizing the evolution of classical symmetric ciphers and their relationship to modern symmetric encryption.